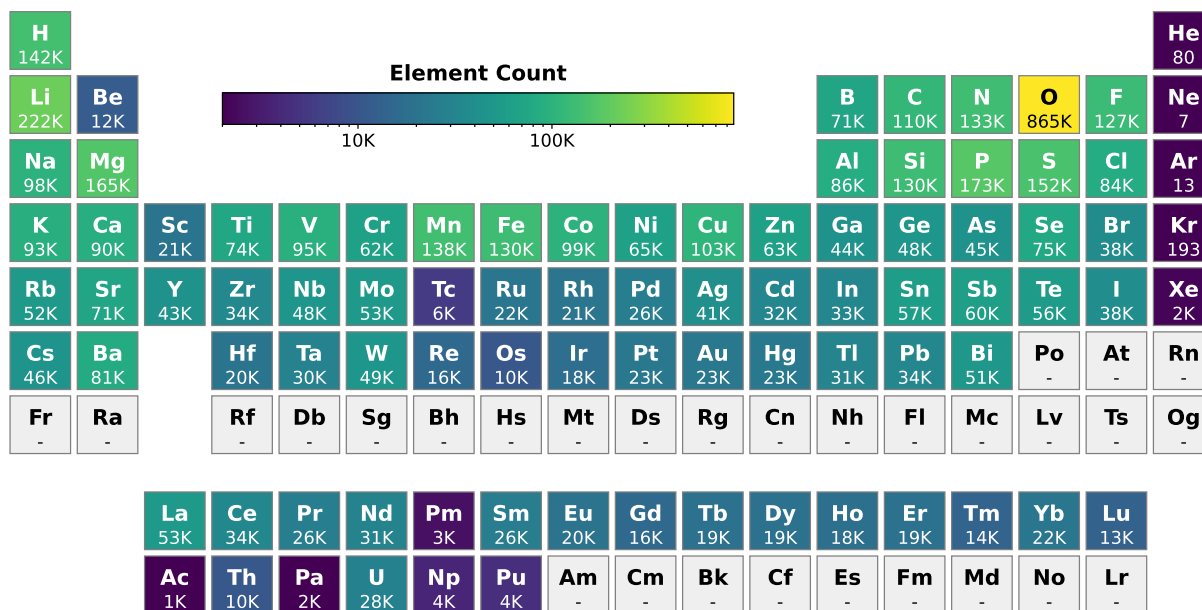
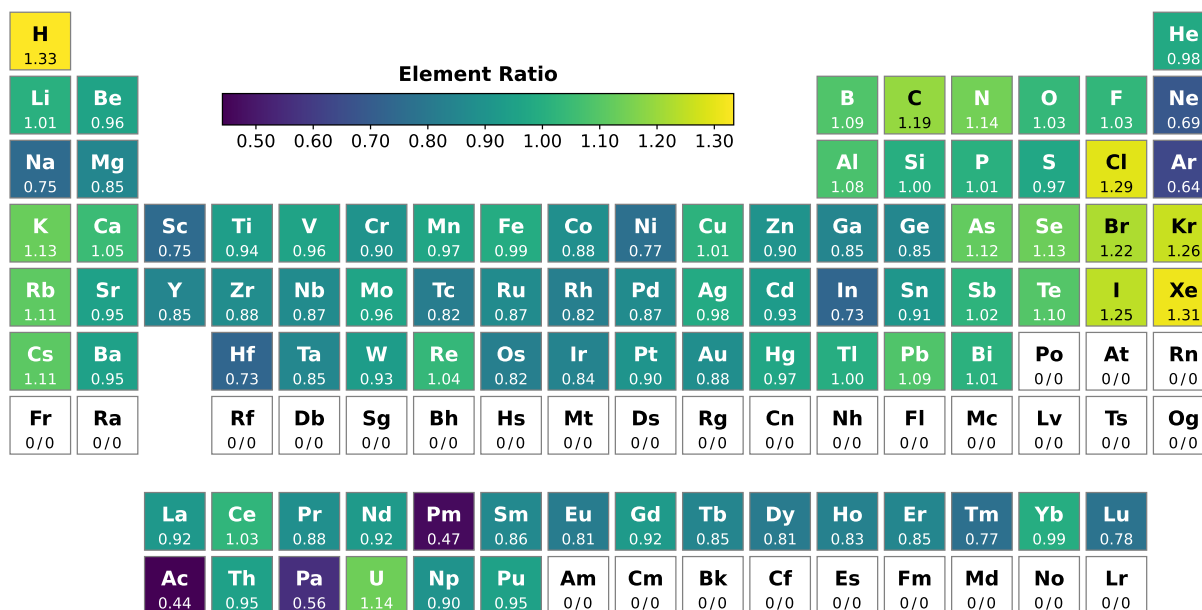


C.3 Exploration of the training data



(a) MPtrj training set element occurrence



(b) Normalized ratio of elements in MPtrj to MP, $\frac{\text{MPtrj} / \text{len}(\text{MPtrj})}{\text{MP} / \text{len}(\text{MP})}$

Figure S66: The number of structures containing a given element in the MPtrj training set (95). MPtrj consists of multiple configurations from every relaxation trajectory in MP. Some elements can require more ionic steps to relax than others. To visualize this, b shows the overabundance of elements relative to the number of structures containing a given element in MP ground states (after normalizing by dividing each dataset by its number of structures). That is, the factor 1.33 for hydrogen in b indicates that structures containing hydrogen were selected 33% more frequently than the base prevalence of hydrogen in MP ground states.

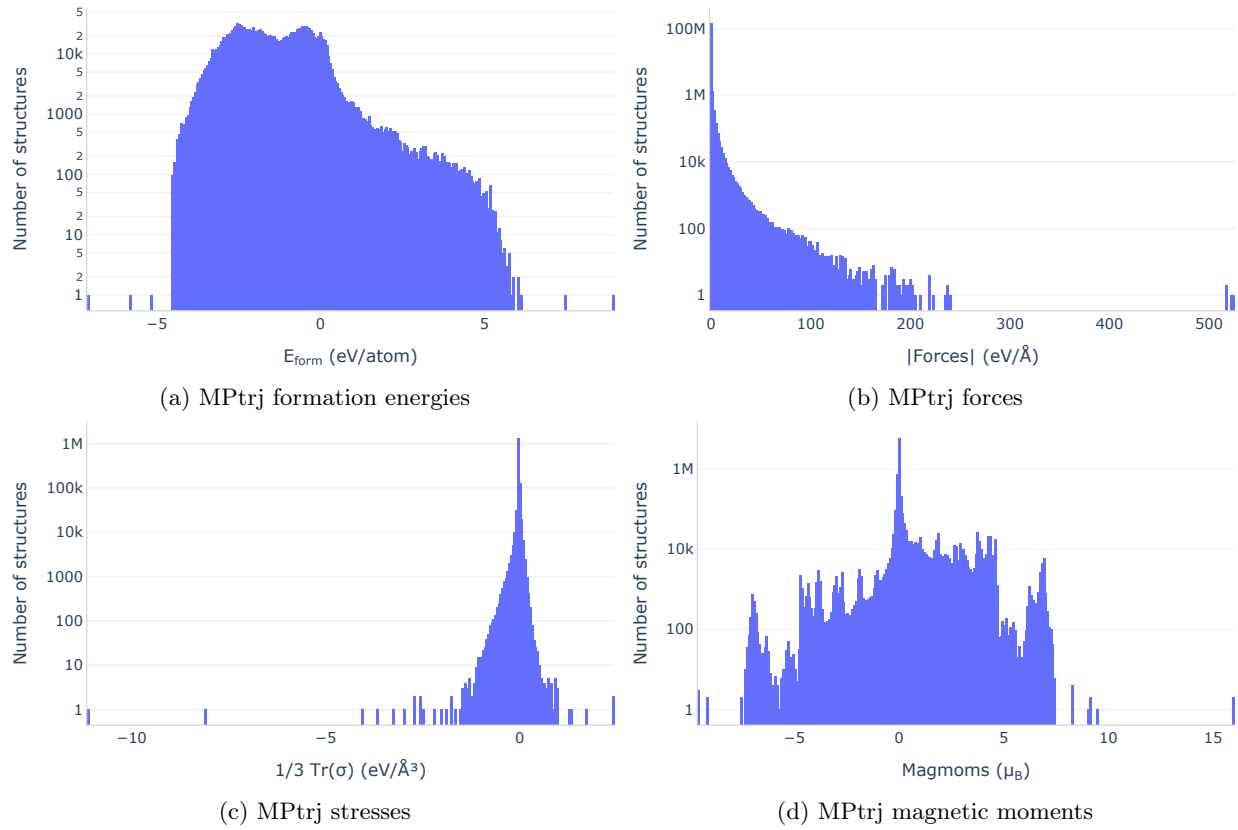


Figure S67: Distribution of energies, forces, stresses and magnetic moments in the MPtrj dataset (95, 280). The bimodality in the formation energy distribution is due to the MP anion correction scheme (325, 326) which significantly lowers oxide formation energies.

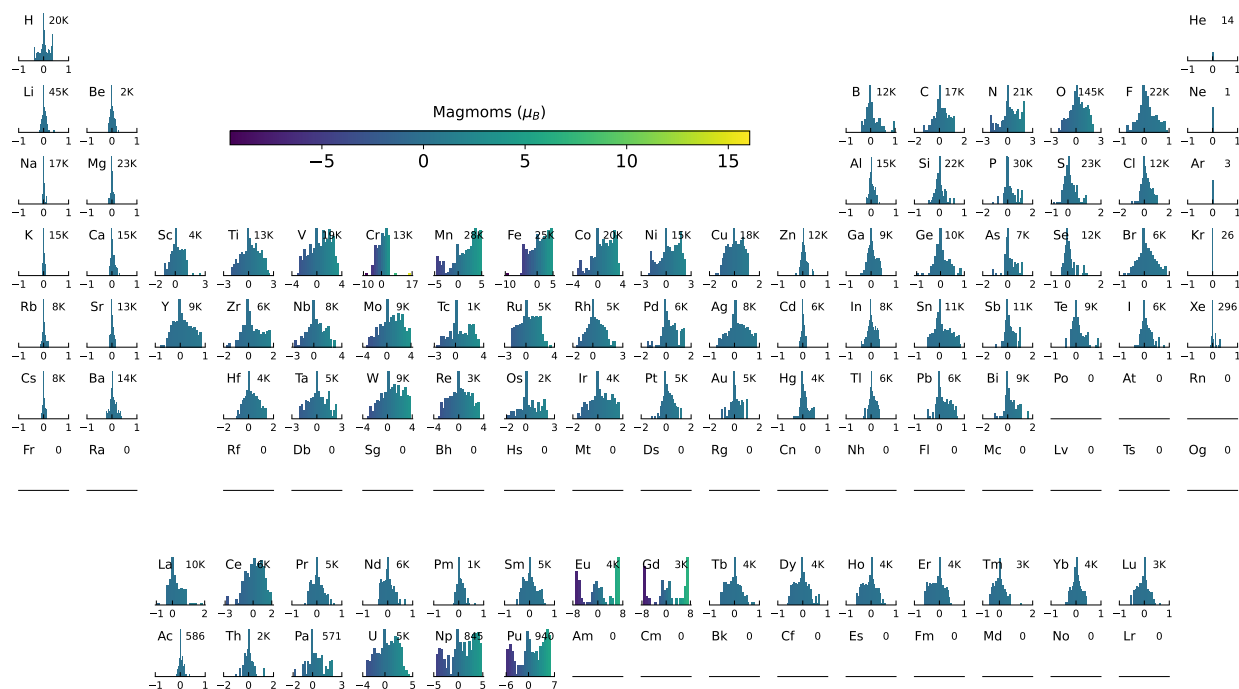


Figure S68: Distribution of magnetic moments for each element in the MPtrj dataset (*95, 280, 327*). The y -axis is log-scaled to allow visualization of the tail of high magnetic moments in some elements with a sharp peak at 0. The number in the top right corner of each element tile counts magnetic moments for that element in the MPtrj dataset. This plot reveals rare erroneous data points in MPtrj. For instance, Cr has a single-point calculation with a highly unphysical magnetic moment of $17\mu_B$.

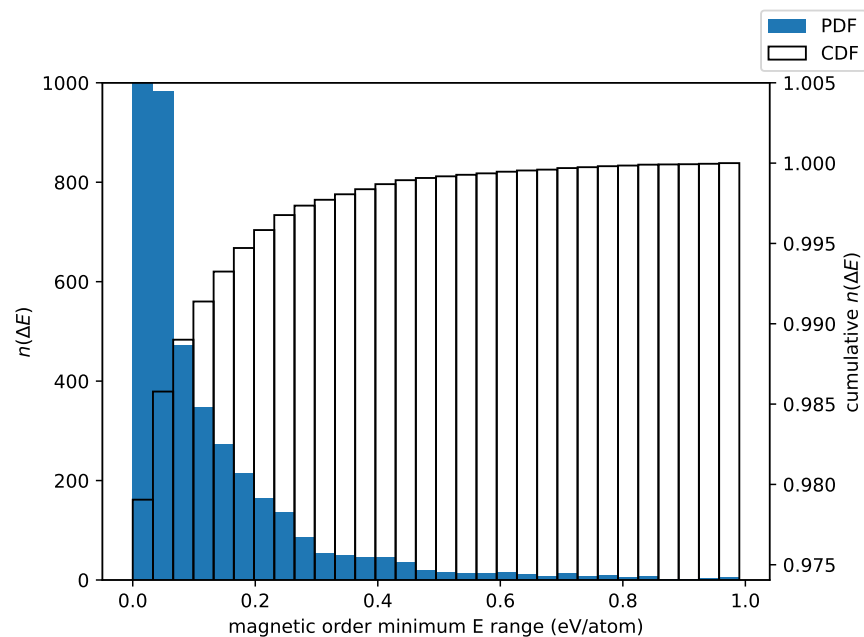


Figure S69: Distribution (PDF) of approximate deviation of energy within each material due to variation in magnetic order (non-magnetic calculation, calculation converges to moment zero, ferromagnetic, and other magnetic orders), with cumulative distribution (CDF) on right axis. The vast majority of materials have very small variation (y axis range does not show full extent of first PDF bar), but a few hundred include different magnetic orders with energies that vary by more than 0.1 eV/atoms

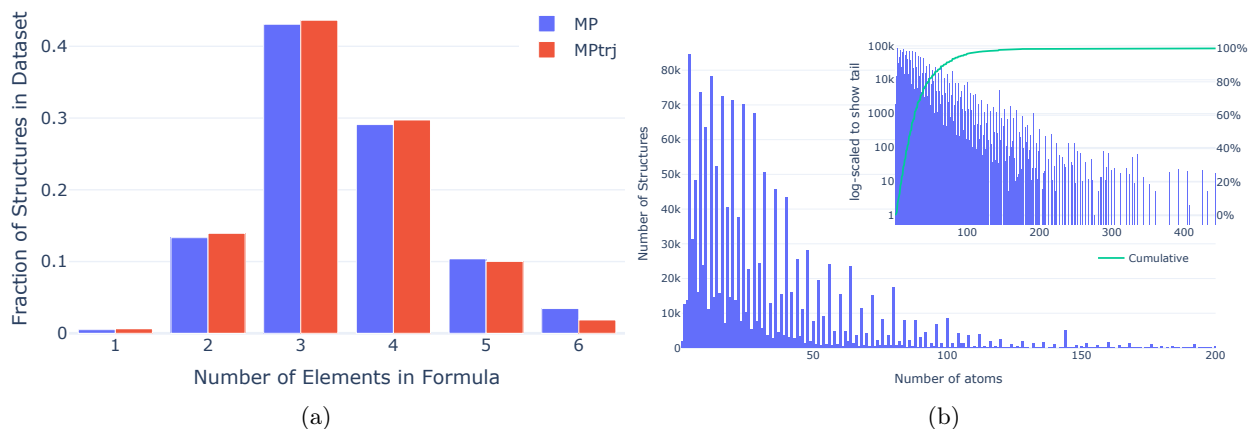


Figure S70: a) Distribution of the number of elements in the compositions of MP structures compared to MPtrj. We observe a slight overabundance of small numbers of elements in MPtrj relative to MP. b) Distribution of a number of sites in MPtrj. The inset shows the same distribution log-scaled to visualize the tail of high site counts. The green cumulative line in the inset shows that 82% have less than 50 sites and 97% of structures in MPtrj have less than 100 atoms.

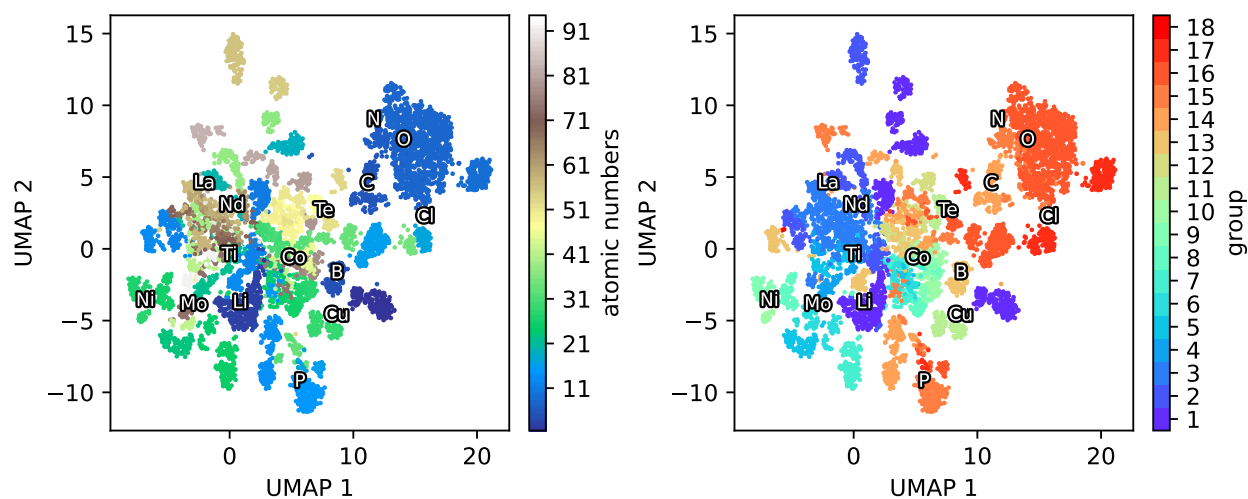


Figure S71: UMAP projection of MACE descriptors for atoms in MPtrj. Each point represents the averaged feature vector of a single element in one structure and is colored by atomic number (left) and group (right) in the periodic table. The features of MACE-MP-0 model are 256-dimensional vectors (concatenation from both first and second layer of 128 channels). Manhattan distance is used for the construction of a high-dimensional UMAP manifold.